Tom Piccolo <tompiccolo50@gmail.com>

I have now fully read your paper- some thoughts

4 messages

Tom Piccolo <tompiccolo50@gmail.com> 8 February 2024 at 10:26
To: Tom Benton <tom.benton@cambridge.org>

Hello Tom

Hope you are well.

I have now read your paper, and I misinterpreted several things and took some statements as literal.

I had some revised points to make:

In the discussion, you do not yourself state that the relation between Single Science and higher PISA scores is causal. **However, you do not exclude it either.** From what I understand, you have left it as a possibility:

So therefore, the relationship could be causal. PISA results could potentially be boosted or increased, but nowhere do you explicitly make that conclusion if everyone was to study Single Science. You leave it open as a possibility.

Hence why you say "we have no idea what would happen if everyone studied Single Science"

So therefore, yes, it is possible that double science students averaging A*s (who did equally well in GCSE's as their high achieving counterparts) could, potentially do as well as their Single Science counterparts.

Their PISA scores could potentially be improved.

GCSE results could potentially increase.

It is important to note that nowhere do you either assert any of the above statements, nor do you exclude any of them.

Also, in line with your research, I have performed calculations and come to the following conclusion:

In 2015, the average PISA Maths score for those students with an A* in GCSE Maths (we are talking about the entire cohort) was 607.

In stark contrast, those with an A* in GCSE Maths but who took Single Science have, approximately, a mean PISA 2015 Maths score of 630-632.

This equates to a 23-25 point score difference.

You have acknowledged that this is correct, that Single Science do better than the rest, but:

But nowhere do you either assert or deny it is causal.

Nowhere do you claim exactly what would happen if everyone studied Single Science.

Both are left open as a possibility.

Many thanks and I hope to receive a response in due course

Tom Piccolo

Tom Benton <tom.benton@cambridge.org> 8 February 2024 at 10:47
To: Tom Piccolo <tompiccolo50@gmail.com>

Hello Tom

There is a world of difference between me saying something and me not excluding it.

No, I don't know for definite that making everyone study separate science whether they want to or not wouldn't increase PISA scores. Neither does anyone else. This does not make it a good idea to assume that their scores would increase.

On another matter, you cannot assume that poor performance of those studying combined science is because they haven't covered the content. You can read chapter 1 of this report (https://www.oecd.org/education/pisa-2015-results-volume-i-9789264266490-en.htm) to work out what is being measured but I think it would be very dangerous to assume that it includes topics not covered in combined science. A lack of curriculum coverage also wouldn't explain why these students also perform better in maths and reading in PISA.

A possibility that has to be considered is that the kinds of students who choose separate sciences are also the kinds of students who are particularly suited to answering the kinds of questions asked in PISA. Perhaps those that enjoy answering short questions of the type asked in PISA rather than creating longer essays or completing practical pieces of work. I don't know this but it's a possibility and not one that can be ruled out just be looking at attainment 8 scores. It's better to be cautious in making conclusions from a set of data.

I consider this conversation closed and will not be replying to any further emails.

Tom

[Quoted text hidden]

Cambridge University Press & Assessment is the publishing and assessment department of the University of Cambridge. The corporate title of the University is *The Chancellor, Masters, and Scholars of the University of Cambridge*. Our principal office is at Shaftesbury Road, Cambridge, CB2 8EA, United Kingdom, with registered VAT number GB 823 8476 09. We are a not-for-profit organisation and an exempt charity under the Charities Act 2011.

Disclaimer
This email is intended solely for the recipient and may contain confidential information. If you have received this email in error, please let us know and then delete the original email and any attachments. Do not use, copy or disclose the contents of this email or any attachment. Any views or opinions in this email are solely those of the author and do not necessarily represent those of Cambridge, unless specifically stated.

Tom Piccolo <tompiccolo50@gmail.com> 8 February 2024 at 13:44
To: Tom Benton <tom.benton@cambridge.org>

Hello

I haven't assumed Double Science misses out content which is in Single Science:

I have noticed several questions in PISA Science on topics such as immunity, homeostasis, fish farming and the nervous system which come up in Single but not Double science.

Also, in PISA Reading many questions require analysis and interpretation of scientific information which is tested heavily in Single Science. And in PISA Maths, specific skill sets which are concentrated in Single Science are also involved.

Also, just to clarify:

You don't think that you can rule out or exclude the notion of PISA scores increasing if Single Science replaces Double?

As I mentioned before, what about those students who did Double Science with equally high GCSE attainment as those who did Single Science? Couldn't their scores theoretically improve? As a potential possibility (I am not saying they would, just a possibility?)

Many thanks

Tom Piccolo
[Quoted text hidden]

Tom Piccolo <tompiccolo50@gmail.com> 26 February 2024 at 01:33
To: Tom Benton <tom.benton@cambridge.org>

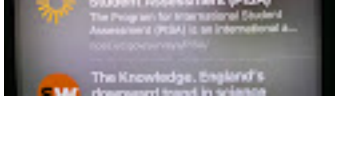
Hello

Just needed some clarification, that's all. I saw an article which spoke of your study online which said "England could rise up PISA rankings if pupils changed GCSE's"

See the attachment

Many thanks

Tom
[Quoted text hidden]



Nine key findings from PISA 2018

PISA 2015: No improvement for a decade ... and 10 other oddit...

England could rise up Pisa table if pupils changed GCSEs, stud...

Program for International Student Assessment (PISA)

The Knowledge. England's

IMG_6214.jpg
2629K